CURRICULUM VITAE

WILLIAM KWESI AKUFFO

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RESEARCH INTERESTS

Multi-agent-robot systems; safety-critical control (control barrier functions); distributed and cooperative model predictive control; learning-based and data-driven MPC; reinforcement learning for control; state estimation under uncertainty (EKF); perception-aware planning and control; convex and numerical optimization; autonomous ground and aerial robots, multi-dof robot manipulation.

EDUCATION

MSc. Mechatronics Engineering — Ashesi University & ETH Zürich (joint MSc/MAS program) Jan 2022 – Dec 2024

Berekuso, Ghana / Zürich, Switzerland

- **Research thesis:** On the Design of Terminal Ingredients for Model Predictive Contouring Control (MPCC) for Autonomous Racing using Reinforcement Learning.
- **Supervisors:** Prof. Dr. Melanie Zeilinger, Dr. Andrea Carron - Institute for Dynamic Systems and Control (IDSC), ETH Zürich.
- Implemented with the ACADOS framework; combines optimal control, learning and numerical optimization.

BSc. Electrical and Electronics Engineering — Ashesi University Sep 2017 – Jun 2021

Berekuso, Ghana

- **Capstone:** A Standalone Inertial Measurement Unit (IMU) Position Determining System for UAVs using Artificial Intelligence. Supervisor: Prof. Dr. Nathan Amanquah.

PUBLICATIONS

[1] **W. K. Akuffo**, O. P. Odekunle, and S. K. Armah, "Transient Response Enhancement in Time-Delayed Teleoperated Systems via Gain Correction," *2024 4th Int. Conf. on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*, Malé, Maldives, 2024, pp. 1–8. doi: 10.1109/ICECCME62383.2024.10797207.

RESEARCH PROJECTS

Learning-Based MPC for Autonomous Racing (MSc thesis, ETH IDSC) 2024

- Designed MPCC terminal ingredients using reinforcement learning to shorten the prediction horizon while retaining closed-loop performance.
- Designed an MPCC warm-start mechanism using the RL-policy in hard-to-guess input regions.
- Extending toward RL-learned, context-conditioned terminal value functions, including a multi-agent variant conditioned on neighboring-agent context.

Vision-Based Pose Estimation for Position-Based Visual Servoing 2026 – present

- Built a real-time 6-DoF pose estimation pipeline on an Intel RealSense D435i (YOLOE segmentation, LightGlue matching, RANSAC) with Kabsch/SVD, projected least-squares.
- Characterizing model-free pose estimators as closed-loop feedback sensors for PBVS and robot arm inverse kinematics, linking perception quality to control performance.
- Applied and characterized numerical solvers for multi-dof robot arm inverse kinematics such as gradient descent, newton raphson, newton's method and gauss-newton

Reconfigurable Dynamical Systems platforms 2026 – present

- Leading the design and development of platforms with module reconfigurable dynamics in one unit.

Autonomous Planters for Large-Scale Farms (Precision Agriculture) Sep 2025 – present

Funded by Prof. Adei Research Studio Grant ($20,000)

- Autonomous traction and navigation unit for a large-scale farm UGV planter.
- Exploring heterogeneous multi-agent interaction between UGV and UAV for optimal cooperative autonomy.
- Hydraulic transmission and propulsion design (Simscape); control strategies for precision seed metering and seed miss compensation.

Modular Autonomous Dynamical-Systems Platforms for Instruction and Research Sep 2025 – present

- Leading development of a 6-DoF robotic arm, an autonomous racing/driving platform, a bicopter, and an EKF-enabled differential-drive rover for crop monitoring, as shared testbeds for control and multi-robot research.

Time-Delayed Teleoperated Robotics Sep 2023 – Dec 2024

- Analyzed stability and transient response of teleoperated systems under communication delay; developed gain correction validated with hardware-in-the-loop DC-motor experiments (publication [1]).

Quadrotor Modelling and Control Dec 2023

- Built UDP-based teleoperation interface software for a quadrotor UAV.
- Performed system identification of the input-to-altitude dynamics for model-based altitude controller design.

ACADEMIC EXPERIENCE

Full-Time Faculty, Department of Engineering — Ashesi University Jan 2025 – present

- Developed and taught Introduction to Engineering (ENGR 112), Mechatronics (ENGR 442) and Advanced Control Systems (ENGR 411) undergraduate courses.
- Supervised five undergraduate capstones (2025 - 2026): a bicopter instructional platform; stereo-vision-based neuromuscular input modelling for prosthetic hand control; a general-purpose 6-DoF robotic arm; an EKF-SLAM-enabled mobile manipulator rover for autonomous crop monitoring; and a modular autonomous driving platform.
- Supervising two undergraduate capstones (2025 - 2027): curvature-aware pure pursuit controller with curvature-aware velocity scheduling for autonomous racing; On Cross-Track Policy Generalization of Actor Critic RL for Autonomous Racing.

Research Assistant, Ashesi Robotics and Controls Lab (ARCLab) Jan 2023 – Aug 2025

- Researched transient-response enhancement and stability of time-delayed teleoperated systems.
- Designed MATLAB/Simulink interface software for DJI Ryze Tello drones, with documentation, and guided students in implementing controllers on the hardware.

Research Assistant, Academic Affairs Research Studio — Ashesi University Jan 2023 – Dec 2024

- Supported PIs with literature reviews, IRB applications, research instruments, data coding and visualization.

INDUSTRY EXPERIENCE

Systems Engineer — Metx Engineering, Accra Jan 2022 – Dec 2022

- Electric-vehicle systems analysis via numerical simulation of design requirements; subsystem integration for product design.

Engineering Intern — OceanLift Subsea, Accra Jul 2024 – Sep 2024

- Designed sea-fastening support structures for offshore cargo mobilization.

Earlier roles 2019 – 2021

- EHS Personnel, Tullow Oil (2021); Software Engineer, Trotro Technologies (2021); Python Programming Tutor, Ashesi (2019).

GRANTS & AWARDS

Prof. Adei Research Studio Grant, Office of the Provost, Ashesi University ($20,000) Sep 2025

Small Research Support, Patrick Awuah Professorial Chair ($4,000) Nov 2025

Ashesi Financial Aid Scholarship, Ashesi–ETH MSc Mechatronics Engineering 2022 – 2024

Winner, Hacklab Hackathon (Team Zetta): AI for Digital Transformation in Africa 2021

TECHNICAL SKILLS

Control & optimization: MPC/MPCC, acados, EKF state estimation, system identification, convex optimization, reinforcement learning, frequency domain, and time domain controller design.

Programming: Python, C, C++, MATLAB/Simulink (incl. Simscape).

Robotics & perception: ROS 2, Intel RealSense, feature matching and 6-DoF pose estimation, hardware-in-the-loop testing, UAV/UGV platform development.

LEADERSHIP & ACTIVITIES

Mechatronics Faculty Lead (2026 – Present), Music Director, The Ensembles, Ashesi (2024) ,Player, Ashesi Sparrows Volleyball (2023–2024)